

An Adaptive Aggregation Scheme for Decentralized Federated Learning in Smart Financial Systems with Non-IID Data

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Abstract. Collaborative machine learning is essential for robust fraud detection in smart financial systems, particularly in UAV-enabled Low-Altitude Economy (LAE) environments. Decentralized Federated Learning (DFL) is thus necessary but suffers under highly heterogeneous (Non-IID) data distributions. We propose DFedADP, a serverless adaptive aggregation scheme. DFedADP integrates decentralized gradient tracking to locally estimate the global update direction. By evaluating the smoothed cosine similarity between local updates and this global direction, the framework dynamically applies a non-linear Gompertz penalty to suppress misaligned updates caused by skewed data. Experiments on Dirichlet-partitioned MNIST [25] and an imbalanced credit card fraud dataset [24] demonstrate that DFedADP achieves superior convergence stability and robustness compared to standard baselines. Under extreme heterogeneity, DFedADP attains 80.55% accuracy, significantly outperforming other DFL methods by 39% to 69%, while approaching the performance of centralized FedAvg (94.58%). On the imbalanced fraud detection task, DFedADP obtains an F1-score of ≈ 0.38 and a Recall of ≈ 0.6 , maintaining strong detection capability without relying on a central server. These results highlight its effectiveness in mitigating client drift and handling data imbalance, making it a practical solution for decentralized, privacy-sensitive financial applications.

Keywords: Decentralized Federated Learning · Adaptive Aggregation · Gradient Tracking · Non-IID · Low-Altitude Economy · Smart Finance

1 Introduction

The rapid expansion of the low-altitude economy enables unmanned aerial vehicles (UAVs) [1] to operate as mobile transaction nodes in smart financial systems. Ensuring secure fraud detection in such environments requires collaborative learning under strict privacy constraints.

However, centralizing data within this LAE environment poses numerous technical and security challenges. Consequently, Federated Learning (FL)[12] emerges as an optimal solution to address these issues. FL enables collaborative model training without sharing raw data, making it well-suited for privacy-sensitive applications. However, its conventional centralized design is unsuitable for UAV networks due to intermittent connectivity, bandwidth limit, and single points of failure [13, 14]. Decentralized Federated Learning (DFL) removes the central coordinator and enables peer-to-peer collaboration. Nevertheless, DFL suffers under heterogeneous (Non-IID) data, where divergent local updates degrade convergence and severely impair global model accuracy. [15, 17].

To overcome this challenge, we propose **DFedADP**, a decentralized adaptive aggregation framework. It integrates gradient tracking with temporally smoothed cosine similarity and a Gompertz-based penalty to suppress misaligned updates and improve robustness under severe data heterogeneity. In summary, our main contributions are as follows:

- We introduce a fully decentralized, privacy-preserving adaptive aggregation framework for federated learning under Non-IID settings.
- We develop an adaptive aggregation mechanism integrating gradient tracking, cosine similarity, and Gompertz penalty.
- We conduct extensive experiments on MNIST[25] and real-world highly imbalanced credit card fraud datasets[24], demonstrating superior convergence stability, robustness, and significantly improved Recall and F1-Score compared to standard decentralized federated learning baselines.

The rest of this paper is structured as follows. Section 2 reviews related work. Section 3 introduces the necessary mathematical preliminaries. Section 4 details our proposed DFedADP framework. Section 5 evaluates its performance, convergence stability, and communication cost under different Non-IID settings comparing to other DFL methods (PSGD, QDFedAvgM). Finally, Section 6 concludes the paper and discusses future research directions.

2 Related Work

2.1 Federated Learning under Data Heterogeneity

Federated learning (FL) enables collaborative model training across distributed clients without requiring the exchange of raw data [12], making it particularly well-suited for privacy-sensitive applications such as financial fraud detection [4, 10]. By keeping data localized, FL reduces privacy risks and regulatory concerns while still leveraging knowledge from multiple participants.

However, the effectiveness of standard FL methods, most notably FedAvg [12], significantly degrades under Non-IID data distributions due to the issue of client drift [7]. In such settings, local updates diverge from the global optimization direction, leading to slow convergence and suboptimal model performance. This problem is further exacerbated in real-world financial systems, where user behavior is inherently diverse and data distributions are highly skewed.

To mitigate these issues, several variants have been proposed. FedProx [15] introduces a proximal term to constrain local updates, while SCAFFOLD [17] employs control variates to reduce variance across clients. FedAdp [18] further adapts aggregation weights based on update characteristics. Although these methods improve convergence stability under heterogeneity, they all rely on a centralized server for coordination and aggregation. This reliance introduces critical limitations, including communication bottlenecks, scalability issues, and vulnerability to single points of failure [9].

These challenges are particularly severe in UAV-enabled edge environments with intermittent connectivity, limited bandwidth, and dynamic topology [1], making a reliable central coordinator impractical. Moreover, financial datasets are highly heterogeneous and imbalanced [2]. Although Focal Loss [6] improves minority-class detection, these methods focus only on loss function design and do not address centralized aggregation limitations or data heterogeneity issues.

2.2 Decentralized Optimization and Gradient Tracking

To overcome the limitations of centralized FL, decentralized federated learning (DFL) has emerged as a promising alternative, enabling direct peer-to-peer collaboration among nodes without requiring a central server [13, 14, 20].

However, most existing decentralized approaches rely on static mixing schemes, such as Metropolis-Hastings weighting [16], to combine information from neighboring nodes. While these methods ensure theoretical convergence under ideal conditions, they fail to account for the varying quality of local updates in heterogeneous environments. As a result, misleading or low-quality updates can propagate through the network, degrading overall performance.

Gradient Tracking (GT) methods [19, 21] address this issue by enabling each node to locally approximate the global gradient through iterative information exchange. By incorporating both consensus and gradient correction terms, GT notably improves convergence stability and reduces the impact of client drift in decentralized settings. Recent works combine GT with adaptive weighting based on update similarity, but integrating these approaches in dynamic, highly Non-IID settings while maintaining efficiency and robustness remains challenging.

3 Preliminaries

3.1 Distributed Optimization Problem

Consider a set of N distributed nodes $\mathcal{V} = \{1, \dots, N\}$, where each node i possesses a private local dataset $\mathcal{D}_i$ with size D_i . Let the total number of samples across all nodes be $D = \sum_{i=1}^N D_i$.

The global learning objective is to find model parameters $w \in \mathbb{R}^d$ that minimize the weighted empirical risk:

$$\min_{w \in \mathbb{R}^d} F(w) \triangleq \sum_{i=1}^N p_i F_i(w) \quad (1)$$

where $p_i = \frac{D_i}{D}$ represents the relative contribution of node i , and $F_i(w)$ denotes the local empirical loss defined over $\mathcal{D}_i$.

3.2 Decentralized Network Topology

The communication structure among nodes is modeled as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where edges indicate bidirectional communication links. Each node i exchanges information only with its immediate neighbors, denoted by $\mathcal{N}_i$.

To facilitate information mixing, we define a doubly stochastic matrix $W \in \mathbb{R}^{N \times N}$, where each entry $W_{ij} > 0$ if $j \in \mathcal{N}_i \cup \{i\}$ and $W_{ij} = 0$ otherwise. The matrix satisfies: $\sum_j W_{ij} = 1$, $\sum_i W_{ij} = 1$ ensuring that both row-wise and column-wise normalization hold. This property guarantees unbiased aggregation and preserves the average of distributed variables. However, it also introduces challenges in information propagation speed and robustness, especially in dynamic or sparse graphs.

3.3 Gradient Tracking and Vector Alignment

To approximate the global gradient in a decentralized setting, each node maintains a local tracking variable $\hat{g}_i^{(t)}$. This variable is iteratively updated using both neighbor information and local gradient differences:

$$\hat{g}_i^{(t+1)} = \sum_{j \in \mathcal{N}_i \cup \{i\}} W_{ij} \hat{g}_j^{(t)} + \nabla F_i(w_i^{(t+1)}) - \nabla F_i(w_i^{(t)}) \quad (2)$$

The update consists of two components: a consensus step that aggregates neighboring estimates, and a correction term that incorporates local gradient changes. Together, these enable $\hat{g}_i^{(t)}$ to track the true global gradient over time, even without centralized coordination.

In addition to gradient tracking, we measure alignment between updates using cosine similarity. Given vectors u and v , the angle is defined as:

$$\theta = \arccos \left(\frac{\langle u, v \rangle}{\|u\| \cdot \|v\|} \right) \quad (3)$$

This metric provides an interpretable measure of directional consistency, where smaller angles indicate better alignment. In our framework, this alignment is crucial for evaluating update reliability and serves as the basis for deriving adaptive aggregation weights that emphasize well-aligned contributions while suppressing divergent ones.

4 Methodology

This section details the proposed DFedADP scheme, which mitigates Non-IID effects among peer nodes by integrating Metropolis-Hastings topology generation, gradient tracking, and an adaptive penalty mechanism.

4.1 Network Topology and Information Exchange

We model the decentralized network as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. At each communication round t , node i computes a stochastic local gradient $g_i(t) = \nabla F_i(w_i(t); \xi_i(t))$. During communication, each node transmits a tuple to its neighbors $i \in \mathcal{N}_j$: $\{w_j(t), g_j(t), \hat{g}_j(t-1)\}$, where $w_j(t)$ denotes local model parameters, $g_j(t)$ the local gradient, and $\hat{g}_j(t-1)$ the previously tracked global gradient estimate.

4.2 Decentralized Global Gradient Tracking

To ensure stable information mixing across the topology, we establish a base doubly stochastic mixing matrix W using the standard Metropolis-Hastings rule, where $W_{ij} = (\max(d_i, d_j) + 1)^{-1}$ for $j \in \mathcal{N}_i$, and d_i is the degree of node i .

In a decentralized environment, the global gradient cannot be accessed directly. Therefore, each node i locally maintains and updates a gradient tracking variable $\hat{g}_i(t)$ to approximate the global gradient:

$$\hat{g}_i(t) = \sum_{j \in \mathcal{N}_i \cup \{i\}} W_{ij} \hat{g}_j(t-1) + g_i(t) - g_i(t-1) \quad (4)$$

The internally tracked variable $\hat{g}_i(t)$ acts as node i 's personal belief of the average update direction of the entire network, serving as a global reference proxy for evaluating incoming updates. This mechanism allows each node to approximate the global gradient direction without requiring centralized coordination.

4.3 Adaptive Consensus Aggregation

With its internally tracked gradient acting as the reference, node i quantifies the contribution level of each incoming update, including its own. Specifically, for each evaluated node $j \in \mathcal{N}_i \cup \{i\}$, node i computes the cosine similarity angle $\theta_{ij}(t)$ between its own tracked global gradient $\hat{g}_i(t)$ and the evaluated local gradient $g_j(t)$:

$$\theta_{ij}(t) = \arccos \left(\frac{\langle \hat{g}_i(t), g_j(t) \rangle}{\|\hat{g}_i(t)\| \cdot \|g_j(t)\|} \right) \quad (5)$$

This provides a principled way to assess the consistency of local updates.

To prevent abrupt weight oscillations caused by noisy local updates, node i applies an Exponential Moving Average (EMA) with a decay rate β to yield a smoothed angle $\tilde{\theta}_{ij}(t) = (1 - \beta)\theta_{ij}(t) + \beta\tilde{\theta}_{ij}(t-1)$.

Next, node i maps this smoothed angle to an adaptive penalty score using a non-linear Gompertz function

$$f(x) = \alpha \left(1 - e^{-e^{-\alpha(x-1)}} \right) \quad (6)$$

which aggressively penalizes neighbors whose updates diverge from node i 's global consensus belief. To prioritize nodes with richer information, this penalty is scaled by the neighbor's dataset size D_j , yielding the adaptive weight factor

$$\phi_{ij}(t) = D_j e^{f(\tilde{\theta}_{ij}(t))} \quad (7)$$

Finally, the consensus aggregation dynamically fuses the topological structure (Metropolis-Hastings base weight W_{ij}) with the computed data-driven adaptive factors. Node i normalizes these consensus weights $\tilde{W}_{ij}(t)$ and aggregates the received models $w_j(t)$ to form its new global model state for round $t + 1$:

$$w_i(t+1) = \sum_{j \in \mathcal{N}_i \cup \{i\}} \tilde{W}_{ij}(t) w_j(t), \quad \text{where} \quad \tilde{W}_{ij}(t) = \frac{W_{ij} \phi_{ij}(t)}{\sum_{k \in \mathcal{N}_i \cup \{i\}} W_{ik} \phi_{ik}(t)} \quad (8)$$

This design enables the system to suppress misleading updates while preserving useful contributions.

As a result, the proposed dual-weighting consensus protocol improves convergence robustness under highly Non-IID data distributions while respecting the decentralized topology. The step-by-step internal workflow of a single UAV node i proceeds as follows:

1. Receiving the incoming message tuples $\{w_j(t), g_j(t), \hat{g}_j(t-1)\}$ from its topological neighbors.
2. Combining these with its local states $\{w_i(t), g_i(t), g_i(t-1), \hat{g}_i(t-1)\}$ to track the global gradient and compute adaptive consensus weights.
3. Producing the updated state $\{w_i(t+1), g_i(t+1), \hat{g}_i(t)\}$ to broadcast in the subsequent communication round.
4. Updating the core inference engine with the newly aggregated parameters $w_i(t+1)$ for real-time transaction processing, ensuring robust fraud detection without exposing raw client data.

To provide a clearer perspective on the node-level execution, Fig. 1 illustrates the step-by-step internal workflow of the DFedADP protocol during a single communication round t , while Algorithm 1 formalizes the overall training procedure. At each round, each node computes its local gradient, exchanges information with neighbors, and updates a local estimate of the global gradient via gradient tracking. Using this reference, it evaluates update alignment through cosine similarity, applies EMA smoothing, and derives adaptive weights via a Gompertz function. The normalized weights are then used to aggregate neighboring models for the next round.

5 Evaluation

We comprehensively evaluate DFedADP under heterogeneous data, examining convergence, robustness to Non-IID settings, fraud detection performance, communication efficiency, and the impact of γ in Focal Loss.

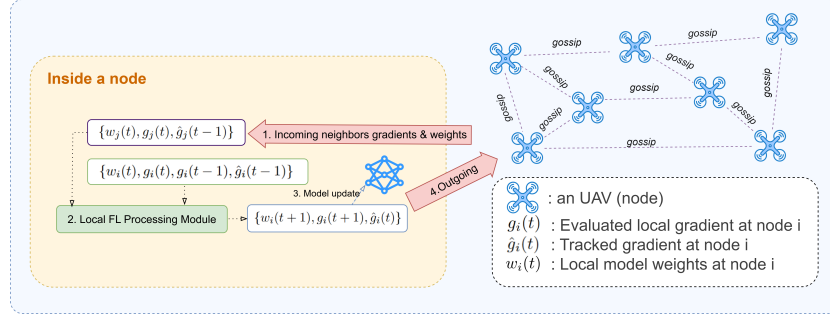


Fig. 1: The overall system architecture of the proposed DFedADP framework within a UAV-enabled decentralized financial system at communication round t . The right panel illustrates the serverless, peer-to-peer graph topology where individual UAVs (nodes) exchange information via a gossip protocol. The left panel illustrates the internal workflow of a single UAV node i .

5.1 Experimental Setup

We evaluate our experiments on a server with dual Intel Xeon Platinum 8462Y+ processors (128 cores), 64 GB RAM, and an NVIDIA H100 GPU under Ubuntu with CUDA 12.6.

MNIST Classification setup We utilize the MNIST [25] dataset; to represent the varying degrees of data heterogeneity, we utilize a Dirichlet distribution to partition the data with concentration parameters $\alpha \in \{10.0, 1.0, 0.5, 0.1\}$, alongside a uniform IID partition. Network sizes are configured with 50 nodes, with a learning rate $\eta = 0.001$. We compare DFedADP against the following baselines: FedAvg [12], PSGD[13], and QFedAvgM [20].

Fraud Detection setup We further evaluate on a publicly available credit card fraud dataset [24], which is highly imbalanced. The model is a multi-layer perceptron with hidden sizes 256–128–64, using Layer Normalization [22], LeakyReLU, and Dropout (0.2). Given the extreme class imbalance where fraudulent transactions constitute a microscopic fraction of the dataset, standard Cross-Entropy loss heavily biases the model towards the majority class (legitimate transactions), rendering standard accuracy metrics deceptive.

To counteract this, we implement a highly customized Focal Loss function [23] to dynamically scale the loss based on prediction confidence:

$$FL(p_t) = -\lambda_t(1 - p_t)^\gamma \log(p_t)$$

where p_t is the model's estimated probability for the target class.

We evaluate the impact of the focusing parameter $\gamma \in \{0.5, 1.5, 2.0\}$. Lower γ (0.5) improves Precision and F1-Score, while higher γ (2.0) increases Recall by emphasizing minority-class examples.

Algorithm 1 DFedADP: Decentralized Adaptive Aggregation

Require: Initial model $w_i(0)$, mixing matrix W , EMA factor β , Gompertz function $f(x) = \alpha(1 - e^{-e^{-\alpha(x-1)}})$

- 1: Initialize $\hat{g}_i(0) = 0$, $g_i(0)$ arbitrary, $\tilde{\theta}_{ij}(0) = 0$
- 2: **for** each communication round $t = 0, 1, \dots, T - 1$ **do**
- 3: $g_i(t) \leftarrow \nabla F_i(w_i(t); \xi_i(t))$ ▷ Local gradient computation
- 4: $\{w_j(t), g_j(t), \hat{g}_j(t-1)\}$ ▷ Receive from neighbors $j \in \mathcal{N}_i$
- 5: $\hat{g}_i(t) \leftarrow \sum_{j \in \mathcal{N}_i \cup \{i\}} W_{ij} \hat{g}_j(t-1) + g_i(t) - g_i(t-1)$ ▷ Gradient tracking
- 6: **for** each $j \in \mathcal{N}_i \cup \{i\}$ **do**
- 7: $\theta_{ij}(t) \leftarrow \arccos\left(\frac{\langle \hat{g}_i(t), g_j(t) \rangle}{\|\hat{g}_i(t)\| \|g_j(t)\|}\right)$ ▷ Cosine angle
- 8: $\tilde{\theta}_{ij}(t) \leftarrow (1 - \beta)\theta_{ij}(t) + \beta\tilde{\theta}_{ij}(t-1)$ ▷ EMA smoothing
- 9: $\phi_{ij}(t) \leftarrow D_j \cdot \exp(f(\tilde{\theta}_{ij}(t)))$ ▷ Gompertz-based weight
- 10: **end for**
- 11: $\tilde{W}_{ij}(t) \leftarrow \frac{W_{ij} \phi_{ij}(t)}{\sum_{k \in \mathcal{N}_i \cup \{i\}} W_{ik} \phi_{ik}(t)}$ ▷ Weight normalization
- 12: $w_i(t+1) \leftarrow \sum_{j \in \mathcal{N}_i \cup \{i\}} \tilde{W}_{ij}(t) w_j(t)$ ▷ Model aggregation
- 13: **end for**
- 14: **return** $w_i(T)$

Given the importance of detecting fraudulent transactions, we set the focusing parameter $\gamma = 2.0$ to down-weight easy examples and apply an extreme weighting factor $\lambda = 0.95$ to heavily penalize misclassifications of the minority (fraud) class, maximizing the Recall metric which is critical in banking security.

5.2 Performance Analysis Across Scenarios

Fig. 2 shows the mean test accuracy and loss over 100 communication rounds on the MNIST [25] dataset. DFedADP consistently outperforms decentralized baselines such as PSGD and QFedAvgM across all heterogeneity levels. Specifically, under the IID setting, DFedADP achieves a final accuracy of 95.71%, successfully surpassing both PSGD (92.90%) and QFedAvgM (82.83%) by absolute performance margins ranging from roughly 3% to 13%.

This superiority becomes highly pronounced under severe Non-IID settings (e.g., $\alpha = 0.1$). Because the global dataset is balanced but spatially skewed across nodes, traditional decentralized methods degrade due to client drift. PSGD completely collapses to 11.39%, and QFedAvgM drops to a sub-optimal 41.66%. In stark contrast, DFedADP maintains stable convergence, preserving a robust high accuracy of 80.55%, thereby outperforming these decentralized baselines by a massive margin ranging from nearly 39% to 69%.

Moreover, DFedADP achieves performance remarkably close to the centralized FedAvg upper bound (which yields 98.02% under IID conditions and 94.58% at $\alpha = 0.1$). This demonstrates its effectiveness in mitigating the adverse impacts of extreme spatial data skewness and preserving global model quality without requiring a central aggregator.

5.3 Performance on Fraud Detection

Figure 3 presents the performance on the imbalanced fraud dataset [24] using Focal Loss. Due to extreme class imbalance, Accuracy becomes unreliable; thus, we focus on Recall and F1-Score.

DFedADP demonstrates strong robustness, significantly outperforming decentralized baselines in overall detection quality. QFedAvgM collapses during training, while PSGD achieve a high Recall of 64.83%, it suffers from poor Precision (24.33%), leading to a degraded F1-Score of 34.37%. In contrast, DFedADP achieves the highest overall F1-Score (≈ 0.38) and competitive Recall (≈ 0.6), closely approaching centralized FedAvg without requiring a central server. This demonstrates that DFedADP effectively balances the trade-off between minimizing false negatives and false positives, surpassing both decentralized and centralized baselines in detection capability without requiring a central aggregator.

5.4 Communication Cost Analysis

As detailed in Table 1, while DFedADP achieves superior accuracy compared to decentralized baselines, it incurs a per-round communication cost of 28.5 MB. This overhead is approximately 96.5% higher than PSGD (14.5 MB) and nearly 295% higher than quantized methods such as QFedAvgM [20] (7.2 MB), primarily due to transmitting full model parameters to ensure high-fidelity aggregation. However, it is worth noting that DFedADP still reduces the communication burden by roughly 35.2% when compared to the 44.0 MB required by central FedAvg. This highlights a clear trade-off between model performance and communication efficiency, making DFedADP less suitable for severely bandwidth-constrained environments despite its robustness.

Table 1: Communication Cost Per Round for Different Algorithms

Algorithm	Cost Per Round (MB)
FedAvg (Central)	~ 44.0
DFedADP	~ 28.5
PSGD	~ 14.5
QFedAvgM	~ 7.2
FedAvg (Clients)	~ 0.9

6 Conclusion and Future Work

In this paper, we introduced DFedADP, a novel Decentralized Federated Learning scheme explicitly designed to tackle the severe data heterogeneity prevalent in smart financial systems. Our empirical evaluations, simulated under various

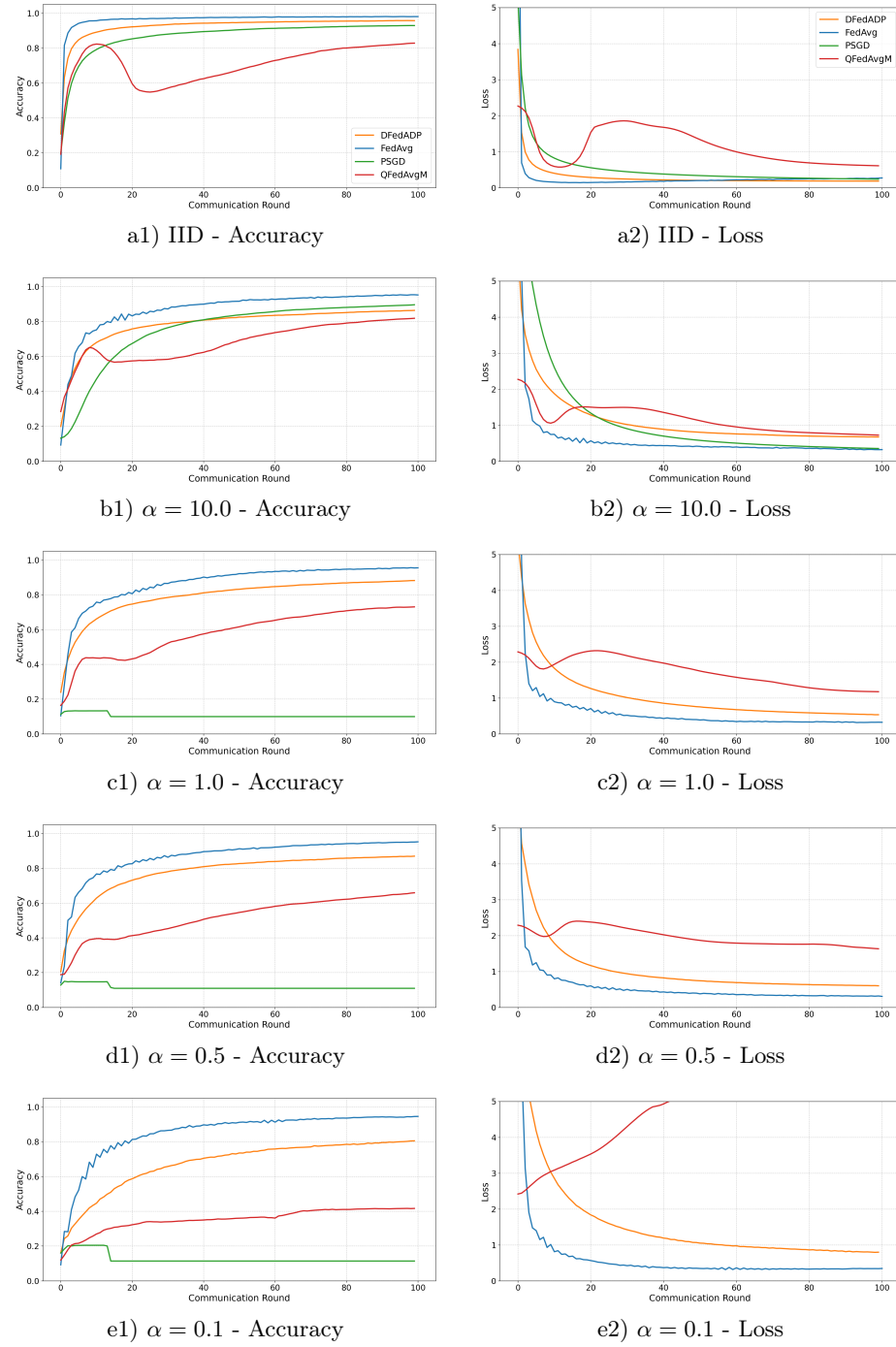


Fig. 2: Mean test accuracy and loss across data heterogeneity scenarios. Each pair (1,2) corresponds to accuracy and loss under the same distribution.

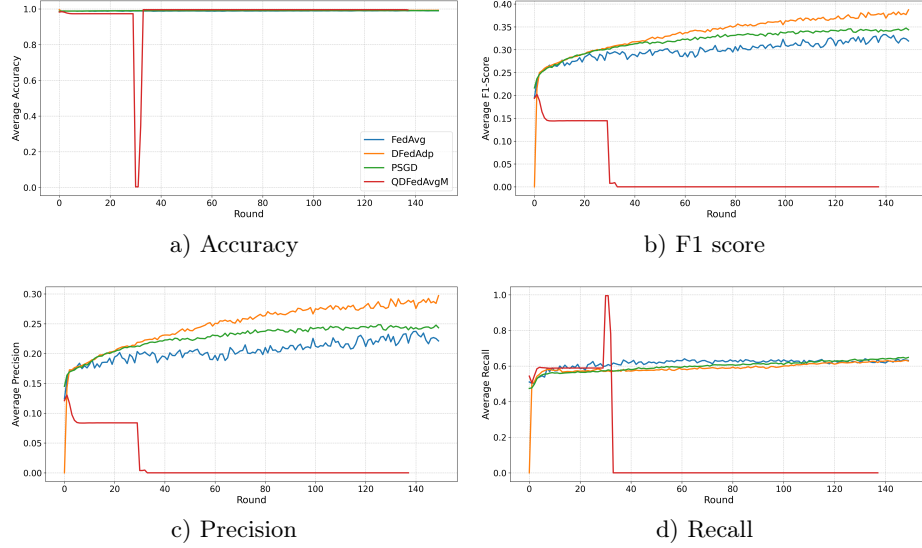


Fig. 3: Performance comparison on an imbalanced credit card fraud dataset using Focal Loss ($\lambda = 0.95, \gamma = 2.0$). DFedADP achieves robust convergence, outperforms decentralized baselines in F1-Score and Recall, and approaches centralized FedAvg performance without a server.

Non-IID settings using Dirichlet distributions, demonstrate that DFedADP significantly outperforms standard decentralized baselines in terms of convergence stability and global model accuracy. It successfully mitigates the adverse impacts of skewed data while strictly preserving institutional privacy without relying on a vulnerable central server.

However, the current formulation of DFedADP incurs a higher communication overhead per round compared to standard methods. Therefore, our future work will primarily focus on integrating communication-efficient techniques, such as model quantization and sparsification, to minimize the bandwidth footprint without sacrificing the adaptive consensus accuracy.

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